

Lifetime quality-adjusted life years lost due to genital herpes acquired in the United States in 2018: a mathematical modeling study



Shiying You,^{a,*} Reza Yaesoubi,^{a,g} Kyueun Lee,^b Yunfei Li,^c Samuel T. Eppink,^d Katherine K. Hsu,^{e,g} Harrell W. Chesson,^d Thomas L. Gift,^d Andrés A. Berruti,^d Joshua A. Salomon,^{f,g} and Minttu M. Rönn^c



^aDepartment of Health Policy and Management, Yale School of Public Health, New Haven, CT, USA

^bDepartment of Health Policy and Management, Graduate School of Public Health, University of Pittsburgh, Pittsburgh, PA, USA

^cDepartment of Global Health and Population, Harvard T.H. Chan School of Public Health, Boston, MA, USA

^dDivision of STD Prevention, Centers for Disease Control and Prevention, Atlanta, GA, USA

^eDivision of Sexually Transmitted Disease Prevention & HIV/AIDS Surveillance, Massachusetts Department of Public Health, Boston, MA, USA

^fCenter for Health Policy / Center for Primary Care and Outcomes Research, Stanford University, Stanford, CA, USA

Summary

Background Genital herpes (GH), caused by herpes simplex virus type 1 and type 2 (HSV-1, HSV-2), is a common sexually transmitted disease associated with adverse health outcomes. Symptoms associated with GH outbreaks can be reduced by antiviral medications, but the infection is incurable and lifelong. In this study, we estimate the long-term health impacts of GH in the United States using quality-adjusted life years (QALYs) lost.

Methods We used probability trees to model the natural history of GH secondary to infection with HSV-1 and HSV-2 among people aged 18–49 years. We modelled the following outcomes to quantify the major causes of health losses following infection: symptomatic herpes outbreaks, psychosocial impacts associated with diagnosis and recurrences, urinary retention caused by sacral radiculitis, aseptic meningitis, Mollaret's meningitis, and neonatal herpes. The model was parameterized based on published literature on the natural history of GH. We summarized losses of health by computing the lifetime number of QALYs lost per genital HSV-1 and HSV-2 infection, and we combined this information with incidence estimates to compute the total lifetime number of QALYs lost due to infections acquired in 2018 in the United States.

Findings We estimated 0.05 (95% uncertainty interval (UI) 0.02–0.08) lifetime QALYs lost per incident GH infection acquired in 2018, equivalent to losing 0.05 years or about 18 days of life for one person with perfect health. The average number of QALYs lost per GH infection due to genital HSV-1 and HSV-2 was 0.01 (95% UI 0.01–0.02) and 0.05 (95% UI 0.02–0.09), respectively. The burden of genital HSV-1 is higher among women, while the burden of HSV-2 is higher among men. QALYs lost per neonatal herpes infection was estimated to be 7.93 (95% UI 6.63–9.19). At the population level, the total estimated lifetime QALYs lost as a result of GH infections acquired in 2018 was 33,100 (95% UI 12,600–67,900) due to GH in adults and 3,140 (95% UI 2,260–4,140) due to neonatal herpes. Results were most sensitive to assumptions on the magnitude of the disutility associated with post-diagnosis psychosocial distress and symptomatic recurrences.

Interpretation GH is associated with substantial health losses in the United States. Results from this study can be used to compare the burden of GH to other diseases, and it provides inputs that may be used in studies on the health impact and cost-effectiveness of interventions that aim to reduce the burden of GH.

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*Corresponding author. Yale School of Public Health, Department of Health Policy and Management, USA.

E-mail address: shiying.you@nyulangone.org (S. You).

^gProfessor.

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Research in context

Evidence before this study

Genital herpes (GH) is the second most prevalent sexually transmitted infection in the United States. Both type 1 (HSV-1) and type 2 (HSV-2) viruses contribute to genital infections in the United States, and HSV-2 accounts for most of the burden of GH. The infection is lifelong, characterized by recurrent and painful blisters during symptomatic outbreaks, leading to substantial health burdens for the general population. Quantifying the overall health burden caused by GH infections is key to evaluating the impact of interventions on GH. We searched PubMed on April 8th, 2022, with terms (HSV OR herpes simplex virus OR genital herpes OR neonatal HSV OR neonatal herpes) AND (quality of life OR QALYs OR quality-adjusted life years OR health impact OR health burden OR DALYs OR disability-adjusted life years) AND (model*). We identified ten studies that evaluated the long-term impacts associated with GH, including seven studies using direct quality-of-life metrics. Among the identified seven studies, six were economic evaluation studies of neonatal herpes examining the benefits of interventions during pregnancy, and the other one evaluated the benefits of screening strategies for both genital and oral HSV. There is a lack of comprehensive quantitative estimates of the health burden of GH across the population and measures which account for the different epidemiology of GH due to HSV-1 and HSV-2.

Added value of this study

We synthesized evidence of adverse health outcomes associated with GH and provided a comprehensive estimation of the burden of GH caused by HSV-1 and HSV-2 by HSV type, sex, age, and neonatal herpes. We developed probability tree models representing the natural history of genital HSV-1 and HSV-2 to estimate the lifetime QALYs lost associated with GH acquired in 2018 in the United States and included

probabilistic sensitivity analyses. We considered adverse outcomes including GH outbreaks, complications, post-diagnosis psychosocial distress, and neonatal herpes. At the population level, the total estimated discounted lifetime QALYs lost associated with adult GH infections acquired in 2018 (excluding neonatal herpes) were 1,930 (95% uncertainty interval (UI) 244–7,330) and 31,200 (95% UI 12,100–62,900) due to genital HSV-1 and HSV-2, respectively. The total estimated burden due to neonatal herpes was 3,140 (95% UI 2,260–4,140) QALYs. At the individual level, the average numbers of QALYs lost was 0.01 (95% UI 0.01–0.02) and 0.05 (95% UI 0.02–0.09) per infection due to genital HSV-1 and HSV-2, and the average QALYs lost associated with per neonatal herpes was 7.93 (95% UI 6.63–9.19). Our estimates indicated the health losses associated with GH among adults were mostly due to the disutility associated with post-diagnosis psychosocial distress and symptomatic recurrences. While the total burden of neonatal herpes in the country is relatively low, the QALYs lost associated with per neonatal herpes infection is significantly higher compared to that associated with per adult GH infection.

Implications of all the available evidence

Our analysis demonstrates the substantial health losses associated with GH infection in the United States. The quantitative quality-of-life estimates can be used to compare the disease burden of GH with other diseases and provides inputs that may be used in health economic studies. The estimated QALYs lost associated with per neonatal herpes compared to per adult GH infection demonstrate the urgent need of developing effective intervention strategies for neonatal herpes. Our study identified gaps in quantifying the health losses associated with psychosocial distress and symptomatic recurrences using standard utility values.

Introduction

Genital herpes (GH), caused by herpes simplex virus type 1 and type 2 (HSV-1, HSV-2), is a common sexually transmitted disease associated with a range of adverse health outcomes (from here on referred to as GH infection).^{1–3} The infection is lifelong, and its natural history is characterized by long asymptomatic intervals and periodic reactivations.³ During symptomatic episodes, GH manifests as painful blisters near the site of infection. GH is also associated with rare complications, such as aseptic meningitis.⁴ Neonatal herpes, an uncommon but devastating complication associated with high morbidity and mortality, can be caused by both HSV-1 and HSV-2.⁵

The estimated number of people aged 18–49 years living with genital HSV in the United States in 2018 was 18 million for HSV-2 and 3–45 million for HSV-1 (assuming 5%–75% of HSV-1 incident infections acquired by people aged 18–49 years old in the United

States are genital).^{6–8} The annual incidence of neonatal herpes in the United States varies between 2.5 and 60 per 100,000 births depending on study years and data sources,⁹ and the Americas region has the highest global rate of the disease.⁵ GH also leads to a substantial monetary burden, especially on families with neonates who develop neurological impairment due to neonatal HSV infections. The direct lifetime medical costs associated with adult GH were estimated at \$972 per treated case,¹⁰ while the lifetime costs of managing neonates with neurological sequelae varied from \$21,000 to \$570,000 per child according to the severity of the condition.^{11–13}

Comprehensive quantitative estimates of the health burden associated with GH are lacking.¹⁴ Previous studies have examined the number of quality-adjusted life years (QALYs) lost due to neonatal herpes, but studies on the number of QALYs lost associated with adult GH are limited.¹⁴ The Global Burden of Disease

(GBD) study estimated that HSV-2 infections would result in 236,000 (95% UI 74,000–554,000) years lived with disability in 2015 globally.¹⁵ These estimates did not account for health losses due to psychosocial distress associated with a GH diagnosis,¹⁶ genitally acquired HSV-1,¹⁷ or neonatal herpes.^{14,17}

We developed probability tree models representing the natural history of HSV-1 and HSV-2 to measure the health losses due to GH. We estimated health losses by computing QALYs lost estimates, which summarize health losses associated with different health states related to GH, including both morbidity and mortality. We estimated the lifetime discounted QALYs lost per GH infection and the total QALYs lost in the US for infections acquired in 2018. We incorporated direct impacts of GH, including health losses to the mother and the child due to neonatal herpes.¹⁸ We did not include indirect effects of GH, such as the increased risk of acquisition of HIV.¹⁹

Methods

Analytic overview

We developed probability tree models to simulate the natural history of genital HSV-1 and HSV-2 (Table 1 and Fig. 1), which captured key features of the natural history of GH in a parsimonious manner, reflected the current evidence-based practices, and achieved computational efficiency. HSV is a life-long infection with painful lesions during symptomatic recurrences and people diagnosed with GH report elevated levels of distress.²⁸ Adverse health outcomes in adults included symptomatic primary and recurrent episodes of GH in a subset of those who acquire infection, psychosocial distress associated with diagnosis and living with GH, mild to moderate complications (urinary retention), and severe complications (aseptic meningitis).^{17,22} We developed a probability tree model to capture the natural history of neonatal herpes, which accounted for health

outcomes after mother-to-child transmission (Fig. 2). The adverse health outcomes for the child included premature mortality and neurological sequelae. For the mother of a child with neonatal herpes, we included quality-of-life losses associated with experiencing the death of a child or the child developing neurological sequelae.¹² Model parameter values governing natural history, testing and treatment, probability and duration of adverse health outcomes, and quality-of-life weights were based on literature reviews (Tables 2 and 3).

We estimated the lifetime number of QALYs lost per incident GH infection acquired in 2018 by age, sex, and HSV type among people aged 18–49 in the US population, and per incident case of neonatal herpes. We conducted probabilistic sensitivity analyses. Details of data sources used to inform model parameterization are provided in Supplement Section S1. All analyses were conducted in Python 3.8 on MacOS using packages including ‘NumPy’, ‘Pandas’, ‘SciPy’, ‘Statsmodels’, and ‘Matplotlib’. The analytic code is available as open access on GitHub: <https://github.com/shiying88/QALYs-Lost-due-to-Genital-Herpes>.

Model of the natural history of genital herpes

The primary infection and recurrent period were the two main classifications of health states in the natural history model for people who acquire genital HSV-1 or HSV-2 infection (Fig. 1 and Table 1). In the model, the primary infection was symptomatic or asymptomatic. Asymptomatic primary infections were more common²² and did not lead to complications during the primary infection period.³⁰ People with symptomatic primary infection could experience mild to moderate complications (urinary retention) or severe complications (aseptic meningitis) in the primary infection period.²⁴ People who experienced recurrences could experience similar mild to moderate complications in the recurrent period, independent of the symptom status of their primary infections.²⁴ For the severe complication, we considered

Primary infection	Characterized by symptomatic viral shedding or asymptomatic viral shedding. ^{20,21} Complications might occur after a symptomatic primary outbreak. ^{4,17,22}
Recurrent period	Recurrence probability is lower among people with asymptomatic primary infections; severity of recurrences is generally lower than that of primary outbreaks. ^{2,23} Complications might occur as result of an outbreak. ⁴
Type of symptomatic recurrence	- No recurrence detected. - Infrequent recurrences. - Frequent recurrences (HSV-2 model only).
Type of sequelae	- Mild to moderate complications after primary infection - Urinary retention due to sacral radiculitis²⁴ - Severe complications after primary infection - Aseptic meningitis²⁴ - Mild to moderate complications after recurrent outbreaks - Urinary retention due to sacral radiculitis²⁴ - Severe complications after recurrent outbreaks - Mollaret's meningitis among people having aseptic meningitis associated with primary HSV (HSV-2 model only)^{5,24–27}

Table 1: Summary of the natural history models of genital HSV-1 and HSV-2 infections in adults.

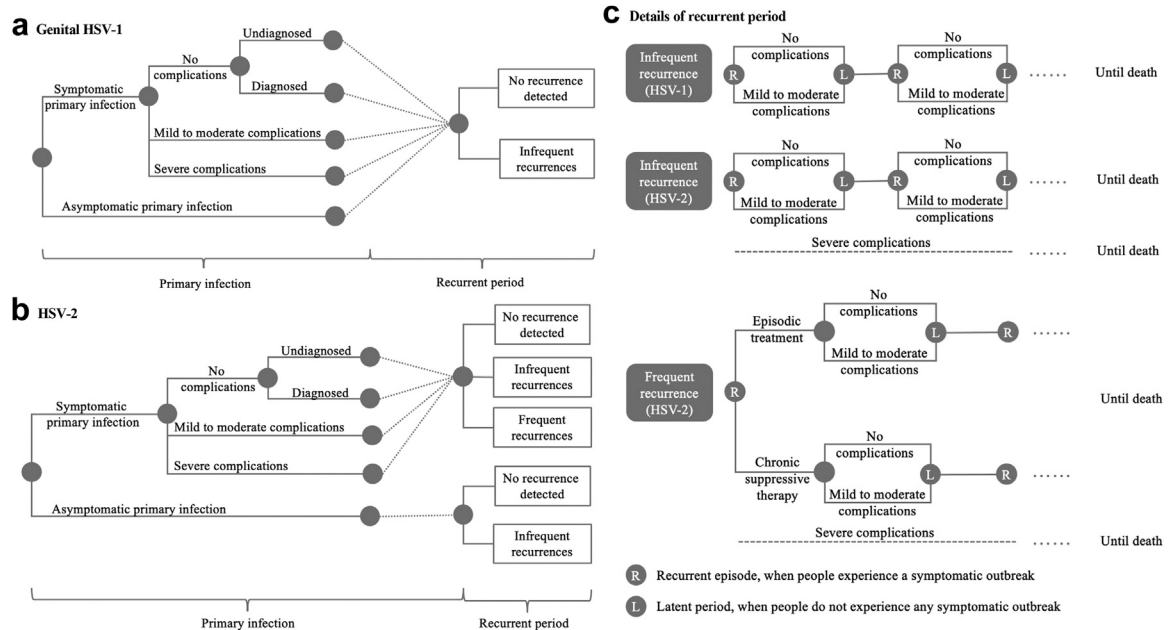


Fig. 1: Probability trees for people with genital HSV-1 (Panel a) and genital HSV-2 (Panel b), and modelled complications following infrequent or frequent recurrences of HSV-1 and HSV-2 outbreaks (Panel c). People with symptomatic primary infection with genital HSV-1 (Panel a) and HSV-2 (Panel b) could experience mild to moderate complications or severe complications in the primary infection period. People with recurrences could also experience mild to moderate complications in the recurrent period (Panel c), regardless of their symptom status during the primary infection period. Severe complications beyond the primary infection stage were possible for i) those with genital HSV-1 upon reaching age 60 years and ii) those with HSV-2 who experienced severe complications in the primary stage. Assumptions regarding the probability, duration, and quality of life impact of complications were based on data regarding urinary retention due to sacral radiculitis (for mild to moderate complications during primary infection and recurrences), data regarding aseptic meningitis (for severe complications during primary infection), and data regarding Mollaret's meningitis (for severe complications beyond primary infection for those with HSV-2).

Mollaret's meningitis among people who had aseptic meningitis with their primary HSV-2 infection, with a probability that was independent of GH recurrences (Supplement Section S2.1; not shown in Fig. 1).^{17,22}

After a primary infection, people might experience recurrent episodes. To estimate the lifetime number of recurrent episodes, we assumed the annual rate of recurrences decreased linearly over time (Supplement Section S3).²⁴ We classified cases into three categories with respect to recurrent episodes: i) No recurrences detected (no symptomatic recurrent episodes detected during the lifetime), ii) infrequent recurrences (initial rate of symptomatic recurrent episodes is less than 6 per year), and iii) frequent recurrences (initial rate of symptomatic recurrent episodes is at least 6 per year).^{17,25} The rate of recurrent episodes was assumed to be dependent on sex, HSV type, and whether the primary infection was symptomatic or asymptomatic (Table 2).

Testing and treatment assumptions

There is no universal screening program for HSV in the US population.²⁶ We assumed people who experience symptomatic outbreaks would seek medical care. The sensitivity of diagnostic testing varied from 80% to 98%, based on the type of tests (PCR or virus culture, with

PCR being more common) and the timing of tests (viral load is higher close to the onset of symptoms and declines thereafter).²⁶ We assumed perfect test specificity. With false negative results, people remained undiagnosed unless experiencing another symptomatic outbreak, at which point they would be tested again. We assumed all people with symptomatic recurrences would be diagnosed within the first year of recurrences.

We considered two types of treatment frequencies: episodic antiviral therapy and chronic suppressive therapy (CST).²² Episodic therapy reduces the severity and duration of the current outbreak without reducing the frequency of future recurrences and associated complications.²⁷ CST requires daily medication, and it reduces viral reactivations and lowers the rate of recurrences compared to people who do not take CST.²⁷ We assumed people with infrequent recurrences opt for episodic treatment over CST. We assumed people with frequent recurrences opt for either CST or episodic treatment; the observed coverage of CST is around 10% among people with frequent recurrences.^{35,27}

Mother-to-child transmission of genital herpes

Neonatal herpes is associated with high morbidity and mortality.⁹ An estimated 50%–80% of affected neonates

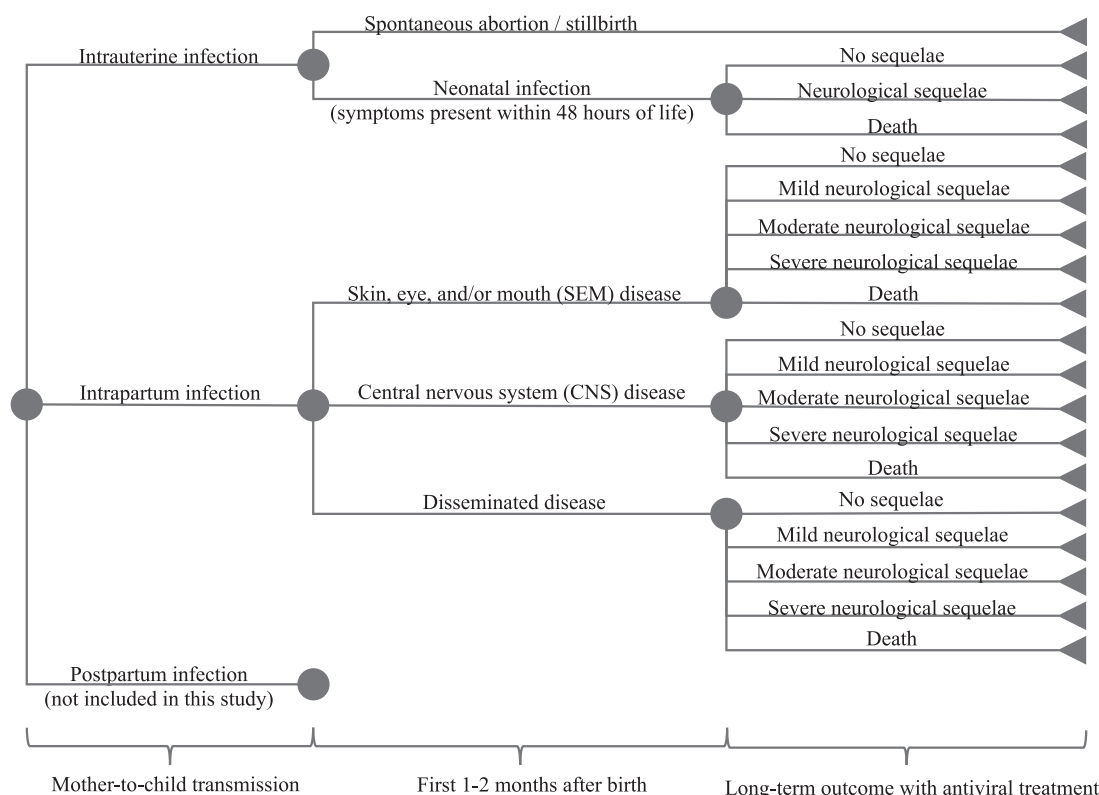


Fig. 2: Probability tree for neonatal HSV infection. The number of QALYs lost due to neonatal HSV is calculated independently from the main HSV-1 and HSV-2 trees. Postpartum infection is generally transmitted through oral contact with caregivers;²⁹ thus, we do not consider this route of infection in our study.

are born to women who acquire genital HSV infection close to term before maternal antibodies have developed.⁹ HSV can be vertically transmitted as intrauterine infection (accounts for 5% of total infections), intrapartum infection (accounts for 85% of total infections), and postpartum infection (accounts for 10% of total infections). We excluded postpartum infection because it is mostly transmitted through oral contact with caregivers who are shedding oral herpes virus commonly caused by HSV-1.^{5,29}

Fig. 2 shows the natural history model for neonatal HSV infection. Intrauterine infections can result in spontaneous abortion, stillbirth, or a neonate showing clinical symptoms within 48 h after birth.⁵³ Three manifestations of the disease have been identified for neonates with intrapartum infection: i) limited outbreaks on the skin, eyes and mouth (SEM), accounting for 45% of the infections, ii) central nervous system (CNS) disease, accounting for 30% of the infections, and iii) disseminated disease, comprising 25% of the infections, where affected neonates show sepsis-like presentations and have multiple organs involved in the dissemination of the virus.^{40,41} One-year morbidity and mortality after intrapartum infection depend on

the primary infection. We divided the outcomes into 5 categories: no sequelae, living with mild neurological sequelae, living with moderate neurological sequelae, living with severe neurological sequelae, and death (descriptive definitions provided in [Supplement Section S5.1](#)).⁵¹ Disseminated infection has the highest mortality, while CNS leads to the highest proportion of neonates living with neurological sequelae.⁴⁰ Given the sparsity of data, we simplified the 1-year outcomes for neonates with intrauterine infection into three categories: no sequelae, neurological sequelae, and death.^{39,54} [Table 3](#) summarizes the point estimates and uncertainty ranges for input parameter values for the natural history of neonatal HSV.

Quality-adjusted life years lost

Estimates of disutility

Disutility estimates were derived primarily from two studies, accounting for reduced health due to infection and impacts of psychosocial distress of infection ([Table 2](#) and [Supplement Section S4](#)). Quality-of-life weights for HSV outbreaks and psychosocial impacts of infection were obtained from a study by Fisman,¹⁸ and quality-of-life weights for complications were

Parameter	Unit	Point Estimates (SD)	Distribution (Shape parameters)	Ref	Parameter	Unit	Point Estimates (SD)	Distribution (Shape parameters)	Ref
Genital HSV-1 infection					HSV-2 infection				
Probabilities					Probabilities				
Primary infection					Primary infection				
Symptomatic outbreak	%	25 (7.5)	Beta (8.08, 24.25)	26,30	Symptomatic outbreak	%	17 (4)	Beta (14.82, 72.37)	24
Proportion of infections diagnosed					Proportion of infections diagnosed				
Symptomatic	%	89 (4.5)	Beta (42.14, 5.21)	26	Same as genital HSV-1 infection				
Asymptomatic	%	0	Fixed	Assumed					
Proportion of diagnosed infections receiving treatment					Proportion of diagnosed infections receiving treatment				
Symptomatic	%	100	Fixed	Assumed	Same as genital HSV-1 infection				
Asymptomatic	%	0	Fixed	Assumed					
Symptomatic recurrent episodes					Symptomatic recurrent episodes				
Type of recurrences after primary symptomatic episode					Type of recurrences after primary symptomatic episode				
None/undetected	%	43	(1 – infrequent)	20	None/undetected	%	11	Dirichlet (N = 203)	31
Infrequent	%	57 (5)	Beta (55.31, 41.73)	20	Infrequent	%	51		31
					Frequent	%	38		31
Type of recurrences after primary asymptomatic episode					Type of recurrences after primary asymptomatic episode				
No/undetected	%	70	(1 – infrequent)	Assumed	None/undetected	%	38	(1 – infrequent)	32
Infrequent	%	30 (5)	Beta (24.9, 58.1)	Assumed	Infrequent	%	62 (5)	Beta (57.81, 35.43)	32
Annual frequency of recurrences (if applicable)					Annual frequency of recurrences (if applicable)				
Men					Infrequent recurrence				
1st year	Rate	1.14 (0.05)	Lognormal (0.13, 0.04)	20	Women	Rate	2.29 (0.05)	Lognormal (0.83, 0.02)	31, calculated
2nd year	Rate	1.00 (0.05)	Lognormal (0.0 0.05)	20	Men	Rate	3.09 (0.05)	Lognormal (1.13, 0.02)	31, calculated
Women					Frequent recurrence (episodic treatment)				
1st year	Rate	2.62 (0.05)	Lognormal (0.96 0.02)	20	Women	Rate	9.87 (0.05)	Lognormal (2.29, 0.01)	31, calculated
2nd year	Rate	2.29 (0.05)	Lognormal (0.83 0.02)	20	Men	Rate	9.76 (0.05)	Lognormal (2.28, 0.01)	31, calculated
					Frequent recurrence (CST)				
					Women	Rate	2.47 (0.05)	Lognormal (0.9, 0.02)	31,33, calculated
					Men	Rate	2.44 (0.05)	Lognormal (0.89, 0.02)	31,33, calculated
Annual avg. reduction in the # of recurrences	Rate	0.5 (0.025)	Lognormal (–0.69 0.05)	34	Annual avg. reduction in the # of recurrences	Rate	0.5 (0.025)	Lognormal (–0.69 0.05)	assumed
					Proportion of people with frequent recurrence taking CST ^a	%	13 (0.39)	Beta (–2.04, 0.03)	35
Complications (during symptomatic primary episode)					Complications (during symptomatic primary episode)				
Both men and women					Men				
Severe complications	%	5	Dirichlet (N = 100)	22	Severe complications	%	1.6	Dirichlet (N = 100)	22
Mild to moderate complications	%	9.25		22	Mild to moderate complications	%	6		22,36, assumed
No complication	%	1 – prob (complications)			No complication	%	1 – prob (complications)		
					Women				

(Table 2 continues on next page)

Parameter	Unit	Point Estimates (SD)	Distribution (Shape parameters)	Ref	Parameter	Unit	Point Estimates (SD)	Distribution (Shape parameters)	Ref
Genital HSV-1 infection					HSV-2 infection				
(Continued from previous page)									
		–			Severe complications	%	6.4	Dirichlet (N = 100)	²²
		–			Mild to moderate complications	%	12.5		³⁶
		–			No complication	%	1 – prob (complications)		
Complications (after each symptomatic recurrent episode)					Complications (after each symptomatic recurrent episode)				
Both men and women					Men				
Mild to moderate complications	%	4.5 (0.25)	Beta (309.38, 6565.62)	²² , assumed	Severe complications ^b	%	30.5 (6)	Beta (17.65, 40.23)	³⁷
		–			Mild to moderate complication	%	3 (1.5)	Beta (3.85, 124.48)	²² , assumed
		–			No complication	%	1 – prob (complications)		
		–			Women				
		–			Severe complications ^b	%	30.5 (6)	Beta (17.65, 40.23)	³⁷
		–			Mild to moderate complications	%	6 (3)	Beta (3.70, 57.97)	²² , assumed
		–			No complication	%	1 – prob (complications)		
Durations					Durations				
Primary symptomatic infection					Primary symptomatic infection				
With treatment	Y ^c	12/365 (1/365)	Lognormal (–3.42, 0.08)	²²	Same as genital HSV-1 infection				
Without treatment	Y	18/365 (2.65/365)	Lognormal (–3.01, 0.15)	²²					
Recurrent symptomatic outbreaks					Recurrent symptomatic outbreaks				
With treatment	Y	7/365 (2.15/365)	Lognormal (–4.00, 0.30)	^{20,22}	With treatment	Y	8.5/365 (1.75/365)	Lognormal (–3.78, 0.2)	^{20,22}
Complications					Complications				
Psychosocial impact	Y	Number of years the person experiences recurrences (Supplement Section S3.3)		^{16,18}					
Mild/moderate complication	Y	5/365 (2.5/365)	Lognormal (–4.40, 0.47)	assumed	Same as genital HSV-1 infection				
Severe complication	Y	8/365 (2.5/365)	Lognormal (–3.87, 0.31)	²²					
Disutilities^d					Disutilities				
Long-term disutility related to diagnosis (psychosocial impact)					Long-term disutility related to diagnosis (psychosocial impact)				
First year of diagnosis	0.01 (0.0025)	Beta (15.83, 1567.17)		¹⁸					
One-time reduction in disutility	0.005 (0.00265) ^e	Beta (3.54, 703.9)		¹⁸	Same as genital HSV-1 infection				

(Table 2 continues on next page)

Parameter	Unit	Point Estimates (SD)	Distribution (Shape parameters)	Ref	Parameter	Unit	Point Estimates (SD)	Distribution (Shape parameters)	Ref
Genital HSV-1 infection					HSV-2 infection				
(Continued from previous page)									
Symptomatic primary and recurrent outbreak (treated and untreated)									
Complications		0.15 (0.038)	Beta (13.09, 74.2)	18	Symptomatic primary and recurrent outbreak (treated and untreated)				
Severe complication		0.05 (0.0125)	Beta (15.15, 287.85)	38	Complications		Same as genital HSV-1 infection		
Mild/moderate complication		0.03 (0.01)	Beta (8.7, 281.3)	assumed			Same as genital HSV-1 infection		
Note: The probabilities and utilities assigned to the outcome of "mild to moderate complications" were based on available estimates of probabilities and utilities for urinary retention. The probabilities and utilities assigned to the outcome of "severe complications" after genital HSV-1/HSV-2 primary outbreak and HSV-2 recurrent outbreak were based on available estimates of probabilities and utilities for primary aseptic meningitis and Mollaret's meningitis, respectively. ^a The proportion of people with frequent recurrences taking chronic suppressive therapy (CST) was estimated based on the disproportionate share of antiviral prescriptions filled around 4 weeks following an initial prescription. ^b See Supplement Section S3 for more details of genital HSV associated severe complications. ^c "y" refers to "year". ^d See Supplement Section S4 for more details of utility estimates. ^e Absolute reduction in estimate after the first year of diagnosis, constant thereafter.									
Table 2: Parameter values and distributions for models of genital HSV-1 and HSV-2 infections among people aged 18–49 years, including natural history of HSV, testing and treatment, duration, and disutilities (utility losses) associated with different health states.									

obtained from the GBD study.⁵⁵ We used a multiplicative model to incorporate background utilities, reflecting chronic comorbidities by age, based on nationally representative EQ-5D index scores.⁵⁶

We included long-term sequelae among neonates. We also accounted for effects on mothers whose children are affected by neonatal herpes, including quality of life losses from losing a child or a child experiencing a mild, moderate, or severe neurological impairment (Table 3 and Supplement Section S5.2). We did not include paternal QALYs lost attributable to neonatal herpes, due to a lack of data in the literature.

Estimates of the incidence of infection

Genital herpes is not a reportable infection, and estimates of incidence rely on seroprevalence data and modelling studies.^{3,6–8} We used estimates of HSV-2 incidence distribution by age and sex for both genital HSV-1 and genital HSV-2 infections (Supplement Fig. S1),^{3,6} given age distributions for genital HSV-1 and HSV-2 prevalence are similar (Supplement Fig. S2).

To estimate the number of children who acquire neonatal herpes, we used the median incidence from a review of neonatal herpes incidence data from local surveillance systems, hospital discharge records, and prospective cohort studies in Canada and the United States.⁹ We excluded 10% of the infections from the total incidence rate, accounting for postnatal infections transmitted through oral contacts.⁹

Table 4 describes the incidence rates and age-specific distributions for genital HSV-1, HSV-2, and neonatal herpes.

Lifetime QALYs lost due to genital HSV-1 and HSV-2 infections in adults

We calculated the average lifetime QALYs lost per incident genital HSV-1 and HSV-2 infection among people aged 18–49 years who acquired infections in 2018. Symptomatic HSV outbreaks and associated psychosocial impacts and complications (mild to moderate, and severe) contributed to the overall QALYs lost, which were computed by summing up the products of the disutilities and durations of events (Supplement Section S3).^{18,55} The adverse maternal health outcomes associated with neonatal herpes were estimated separately in the section below, and they were not included in the total QALYs lost associated with adult GH infections. QALYs lost were computed using a life table approach, which tracked survivorship and health losses in a cohort exposed to age-specific mortality rates. Future health losses associated with current infections were discounted to the year of infection using a discount rate of 3% per year.⁵⁶

We estimated the total expected lifetime QALYs lost due to GH infections acquired in the United States in 2018.⁶ We assumed 20% of HSV-1 incident infections and all HSV-2 infections are at the genital site for the

Parameter	Unit	Point Estimates (SD)	Distribution (Shape parameters)	Ref
Probabilities				
Type of HSV neonatal infection				
Intrauterine infection	%	5	Dirichlet (N = 500)	5
Intrapartum infection	%	85		5
Postpartum infection (excluded)	%	10		5
Intrauterine infection outcomes				
Death before birth	%	7.8 (1.5)	Beta (24.85, 293.77)	39
Short-term manifestations after birth (within 48 h)^a				
Only skin, eye disease (SE)	%	32 (5.8)	Beta (20.38, 43.31)	39
Central nerves system disease	%	68	1-prob(SE)	39
12-month outcomes after infection (all assumed to receive treatments)				
No sequelae	%	16.0	Dirichlet (N = 51)	39
Neurological sequelae	%	25.0		39
Death	%	59.0		39
Intrapartum infection outcomes				
Short-term manifestations after birth				
Skin, eye, and/or mouth (SEM) disease	%	45	Dirichlet (N = 200)	40
Central nervous system (CNS) disease	%	30		40
Disseminated disease	%	25		40
12-months outcome after infection (all assumed to receive treatments)^b				
Disseminated disease				
No neurological sequelae	%	59	Dirichlet (N = 200)	13,41,42
Mild neurological sequelae	%	1		13,41-44, assumed
Moderate neurological sequelae	%	3		13,41-44, assumed
Severe neurological sequelae	%	8		13,41-44, assumed
Death	%	29		13,41,42
CNS disease				
No neurological sequelae	%	30	Dirichlet (N = 200)	13,41,42
Mild neurological sequelae	%	13		13,41-44, assumed
Moderate neurological sequelae	%	17		13,41-44, assumed
Severe neurological sequelae	%	36		13,41-44, assumed
Death	%	4		13,41,42
SEM disease				
No neurological sequelae	%	98	Dirichlet (N = 200)	13,41,42
Mild neurological sequelae	%	1		13,41-43, assumed
Moderate neurological sequelae	%	1		13,41-43, assumed
Severe neurological sequelae	%	0		13,41-43, assumed
Death	%	0		13,41,42
Durations				
Maternal				
Losing a child (initial period)	Y ^c	0.6 (0.475)	Lognormal (-0.75, 0.7)	45,46, assumed
Losing a child (after initial period)	Y		Maternal life expectancy-initial period (Supplement Section S5.2)	12,47,48, assumed
Child's mild neurological sequelae	Y	53.2 (1)	Lognormal (3.97, 0.02)	12,48
Child's moderate neurological sequelae (initial period)	Y	1 (0.5)	Lognormal (-0.11, 0.47)	12,48, assumed
Child's moderate neurological sequelae (after initial period)	Y		Maternal life expectancy - initial period (Supplement Section S5.2)	12,48, assumed
Child's severe neurological sequelae	Y	20 (10)	Lognormal (2.88, 0.47)	12
Neonatal				
Child's mild to moderate neurological sequelae	Y	78.8	Fixed	12,48
Child's severe neurological sequelae	Y	20 (10)	Lognormal (2.88, 0.47)	12

(Table 3 continues on next page)

Parameter	Unit	Point Estimates (SD)	Distribution (Shape parameters)	Ref
(Continued from previous page)				
Life expectancies				
Maternal				
Average age of pregnancy	Y	29		⁴⁹
Life expectancy (at 29 years old)	Y	53.2 (1)	Lognormal (3.97, 0.02)	⁴⁸
Neonatal				
No sequelae	Y	78.8	Fixed	⁴⁸
Mild/Moderate neurological sequelae	Y	78.8	Fixed	⁴⁸
Severe neurological sequelae	Y	20 (10)	Lognormal (2.88, 0.47)	¹²
Disutilities				
Maternal				
Losing a child (initial period)		0.08 (0.035)	Beta (4.73, 54.36)	^{12,47,50}
Losing a child (after initial period)		0.06 (0.025)	Beta (5.35, 83.89)	^{12,47,50} , assumed
Child's neurological sequelae				
Mild		0.06 (0.025)	Beta (5.35, 83.89)	¹²
Moderate (initial period)		0.13 (0.025)	Beta (23.39, 156.57)	¹²
Moderate (after initial period)		0.06 (0.025)	Beta (5.35, 83.89)	¹² , assumed
Severe		0.24 (0.07)	Beta (8.69, 27.53)	¹²
Neonatal (intrauterine infection)				
No sequelae		0.0	Fixed	
Neurological sequelae		0.38 (0.1075)	Beta (7.37, 12.02)	⁵¹ , assumed
Death		1.0	Fixed	
Neonatal (intrapartum infection)				
No sequelae		0.00	Fixed	⁵²
Mild neurological sequelae		0.18 (0.05)	Beta (10.45, 47.59)	⁵²
Moderate neurological sequelae		0.48 (0.05)	Beta (47.44, 51.4)	⁵²
Severe neurological sequelae		0.84 (0.05)	Beta (44.32, 8.44)	⁵²
Death		1.00	Fixed	⁵²

^aNo disseminated disease observed in this study, potentially because such widespread infection can lead to spontaneous abortion or stillbirth before delivery. ^bCriteria for sequelae categorization can be found in [Supplement Section S5.1](#). ^c'Y' refers to 'year'.

Table 3: Parameter values and distributions for model of neonatal HSV infection, including natural history of HSV, testing and treatment, duration, and disutilities (utility losses) associated with different health states.

base case,⁸ and included an uncertainty range based on Spicknall et al. (2021) to capture the uncertainty around the incidence of genital HSV-1.⁶

Lifetime QALYs lost due to neonatal herpes

The average lifetime QALYs lost per neonatal herpes infection accounted for health losses to children with neonatal herpes and those experienced by their mothers. For neonates, the health loss for a premature death due to HSV was equal to the discounted total quality-adjusted life years of a healthy child without neonatal herpes. We estimated the long-term QALYs lost for children with neurological sequelae, which varied based on the severity of the condition and the life expectancy of the child with the condition. For mothers of children with neonatal herpes, we accounted for lifelong effects of the death of a child, with a disutility of 0.08 assumed for the first 0.6 years after the child's death and a disutility of 0.06 thereafter. We assumed that neurological sequelae caused by neonatal herpes

would have an impact on the mother's quality of life for the duration of the mother's life expectancy or the child's life expectancy, whichever was shorter.^{12,47} We assumed the maternal disutilities associated with child's mild and severe neurological sequelae were constant, while the maternal disutility associated with child's moderate neurological sequelae was 0.13 during the first year of neurological sequelae and decreased to 0.06 thereafter ([Supplement Section S5.2](#)). The average age of pregnancy was estimated to be 29 years, based on the birth rates across age groups in 2018 in the US.⁴⁹

We estimated the average lifetime QALYs lost per neonatal herpes infection and the total lifetime QALYs lost due to neonatal herpes acquired in the US in 2018.

Sensitivity analyses

We performed a probabilistic sensitivity analysis. We accounted for the combined effect of uncertainties in all input parameters. We sampled parameters 1,000 times independently using pre-defined distributions ([Tables 2,](#)

Parameter	Unit	Point Estimates (SD)	Distribution (Shape parameters)	Ref
<u>Incidence of genital HSV-1 (rounded to thousands)</u>				
Men and women		140,000 (125,000)	Lognormal (11.56, 0.77)	6,8
Proportion of male genital HSV-1	%	51 (5)	Beta (50.47, 48.49)	3, assumed
<u>Incidence of genital HSV-2 by sex (rounded to thousands)</u>				
Men		322,000 (121,000)	Lognormal (12.62, 0.36)	6
Women		275,000 (100,000)	Lognormal (12.46, 0.35)	6
<u>Age distribution of genital HSV-1 and HSV-2 incidence</u>				
Men				
18–24 years old	%	30.0	Dirichlet (N = 1000)	3,6
25–29 years old	%	23.2		3,6
30–34 years old	%	11.2		3,6
35–49 years old	%	35.6		3,6
Women				
18–24 years old	%	54.0	Dirichlet (N = 1000)	3,6
25–29 years old	%	21.9		3,6
30–34 years old	%	7.4		3,6
35–49 years old	%	16.7		3,6
<u>Incidence rate for neonatal infection (2018)</u>				
Incidence rate of neonatal infection		11.5 (1.5) per 100,000 live births	Lognormal (2.43, 0.13)	5,9,49

Table 4: Incidence and age-specific distributions for genital HSV-1 and HSV-2 among people aged 18–49 years old, and neonatal herpes in the United States in 2018.

Table 4: Incidence and age-specific distributions for genital HSV-1 and HSV-2 among people aged 18–49 years old, and neonatal herpes in the United States in 2018.

3, and 4). The standard deviations for pre-defined distributions were approximated as one-fourth of the 95% uncertainty ranges of the parameters. Using the model outputs derived from 1,000 simulations, we present mean estimates and 95% uncertainty intervals.

In addition, we performed one-way sensitivity analyses for variables that have very limited sources to support parameterization but which may significantly influence the QALYs lost estimates. The one-way sensitivity analyses included the psychosocial loss due to HSV diagnosis¹⁸ and the utility weight for a symptomatic HSV outbreak.^{18,55} We also examined the sensitivity of the results to the choice of discount rate.

Role of the funding source

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Results

Average QALYs lost per infection in the adult population

We estimated the number of QALYs lost per adult GH infection to be 0.05 (95% UI 0.02–0.08), equivalent to losing 0.05 years or about 18 days of life for one person

with perfect health. The average number of QALYs lost per GH infection due to genital HSV-1 and HSV-2 was 0.01 (95% UI 0.01–0.02) and 0.05 (95% UI 0.02–0.09), respectively. Fig. 3 presents the age and sex-specific QALYs lost per genital HSV-1 and HSV-2 infections. Estimates were similar across age groups, as we assumed the rate of symptomatic recurrences and most of the disutilities were independent of age. Recurrences declined over time, and people living with HSV-2 experienced a higher rate of recurrences than people with genital HSV-1, leading to a higher total QALYs lost per infection caused by HSV-2. Women were estimated to have a slightly higher loss due to genital HSV-1 compared to men while men had a higher loss from an HSV-2 infection compared to women. These differences arose because women with genital HSV-1 had a higher recurrence rate,²⁰ and men with HSV-2 experienced a higher recurrence rate on average than women with HSV-2.³¹ The maternal health losses associated with neonatal herpes was modelled separately and reported in the section of QALYs lost due to neonatal herpes.

Fig. 4 shows the breakdown of QALYs lost per GH infection attributed to the primary outbreak, recurrent outbreaks, psychosocial impacts, mild to moderate complications, and severe complications. Estimated health losses were mostly due to recurrences, including the total number of symptomatic recurrences and the long-term psychosocial impacts.

Total QALYs lost in 2018 in subpopulations and the entire population

The total QALYs lost in the general population and by sex are presented in Fig. 5. For the US population aged

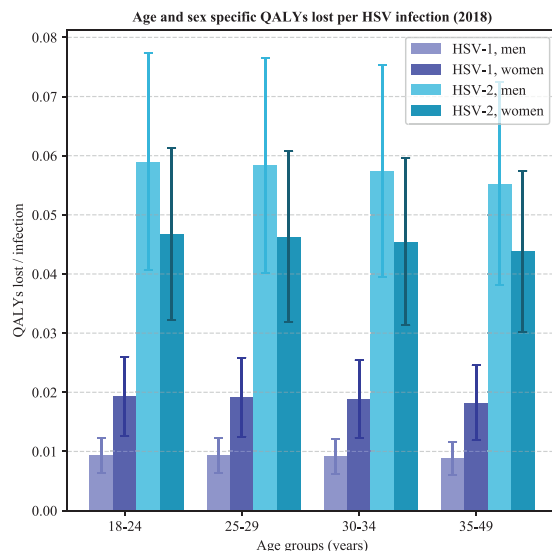


Fig. 3: Average number of QALYs lost per GH infection acquired in 2018 among people aged 18–49 years in the U.S. population, by age group, sex, and HSV type. Error bars represent one standard deviation uncertainty below and above point estimates, which were approximated as one-fourth of the 95% uncertainty ranges. The precise results are reported in the Supplement (Suppl. Table S2).

18–49 years, there were an estimated 137,000 (95% UI 22,300–435,000) incident genital HSV-1 infections and 598,000 (95% UI 347,000–962,000) HSV-2 infections,^{6–8} which resulted in 1,930 (95% UI 244–7,330) and 31,200 (95% UI 12,100–62,900) lifetime QALYs lost due to genital HSV-1 and HSV-2, respectively.

QALYs lost due to neonatal herpes

The overall lifetime QALYs lost per neonatal infection was estimated to be 7.93 (95% UI 6.63–9.19). This estimate included QALYs lost for mothers, estimated to be 0.77 (95% UI 0.42–1.22) per neonatal infection (Fig. 6). In 2018, the estimated number of neonatal herpes infections due to GH was 396 (95% UI 303–499), which corresponded to an estimated total burden of neonatal HSV infections in 2018 of 3,140 (95% UI 2,260–4,140) QALYs.

Sensitivity analyses

The probabilistic sensitivity analyses are reflected in the 95% UI of the main analyses. Fig. 7 presents the results of one-way sensitivity analyses. Results were sensitive to assumptions on the amount of psychosocial disutility and the disutility associated with symptomatic outbreaks. Assuming a lower disutility for symptomatic outbreaks (based on the estimates in the GBD study)⁵⁵ resulted in lower QALYs lost estimates, and the reduction was more pronounced for HSV-2. The number of QALYs lost per infection was lowered by approximately 40% in absence of psychosocial distress, and increased

by 50% when we doubled the psychosocial impacts to 0.02. The change in QALYs lost between varying discounting rate for genital HSV-1 was minor, mainly attributed to the fact that the recurrent outbreaks were mostly seen within the five years after the primary episode and there was no other long-term adverse outcome.

Discussion

We estimated the burden of genital herpes in the US population, including the average lifetime number of QALYs lost per infection and the total number of QALYs lost due to GH infections acquired in 2018. The estimates allow for more comprehensive estimates of the burden of genital herpes in the United States beyond estimates of incidence and prevalence alone. The estimates enable comparison of disease burdens across various STIs and other adverse health conditions. Our findings can inform health economic studies, such as resource allocation models and cost-effectiveness analyses of interventions to prevent adult GH and/or neonatal herpes.¹⁴ Our estimates reflect the average number of QALYs lost associated with per GH infection, consisting of both symptomatic and asymptomatic GH infections. Had we excluded the asymptomatic infections and instead estimated the number of QALYs lost per symptomatic, diagnosed GH case, the resulting estimate would have been notably higher.

Consistent with previous studies, we estimated that HSV-2 accounts for the majority of the burden of GH.³ The seroprevalence of HSV-2 peaked at 19.9% in 1983, and despite the general declining trend in HSV-2 seroprevalence and incidence, a high incidence is projected to persist for at least the next three decades.⁷ Taking a global perspective, while the African region contributes to the highest portion of the burden of genital ulcer disease (GUD) due to HSV-2, the burden of GUD in the Americas is higher than that in Asia and Europe.⁵⁷ The number of person-days with GUD due to genital HSV-1 was also estimated to be the highest in the Americas, accounting for nearly half of all genital-HSV-1 cases globally (42 million out of 92 million GUD person-days). We quantified the total QALYs lost associated with both genital HSV-1 and HSV-2, and also considered neonatal herpes.^{14,15}

Our estimates can be used to compare the burden of GH infection to other STIs and infections. On a per-infection basis, the estimated average number of QALYs lost due to GH (0.05 QALYs) is smaller than that due to HIV (5.80 QALYs),^{58,59} syphilis (0.09 QALYs),⁶⁰ and gonorrhoea (among women, 0.09 QALYs),⁶¹ while the burden of GH is higher than that of per gonorrhoea infection among men (0.0015–0.002 QALYs)⁶¹ and per case of genital warts (0.024 QALYs).^{62–64} At the population level, the estimated total quality-of-life burden due to GH in the US (33,100 QALYs) exceeds that of syphilis (13,300 QALYs),⁶⁰ mainly due to the higher incidence of GH.

Breakdowns of QALYs lost per HSV infection by sex and viral type

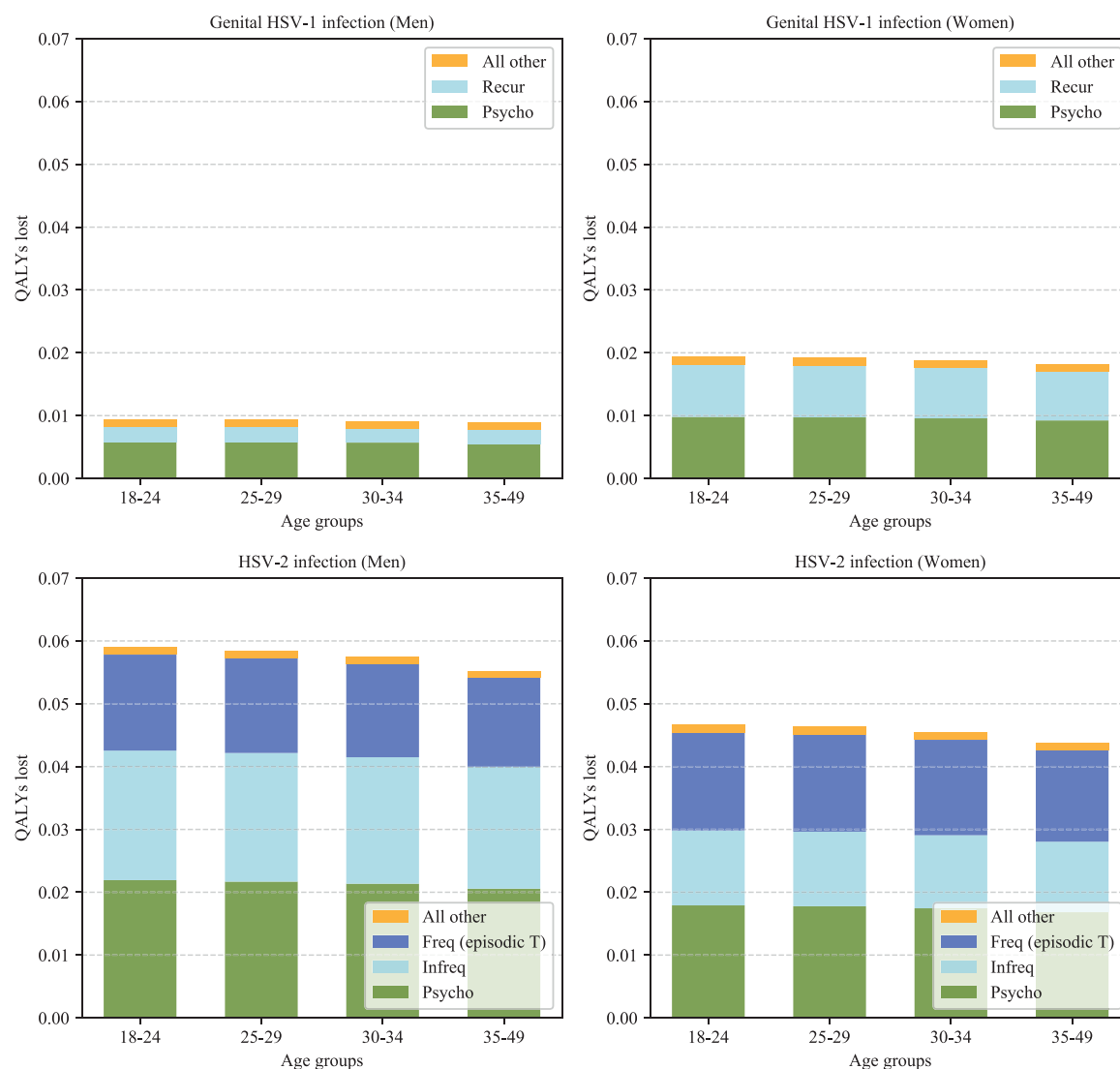


Fig. 4: Breakdowns of QALYs lost per GH infection acquired in 2018 for people aged 18–49 years in the U.S. population. The upper graphs show the percentage QALYs lost per genital HIV-1 infection that can be attributed to recurrent outbreaks ('Recur'), psychosocial impacts ('Psycho'), and 'all other' (including the primary outbreak, mild to moderate complications, and severe complications). The lower graphs show the percentage QALYs lost per HSV-2 infection that can be attributed to the psychosocial impacts ('Psycho'), infrequent recurrent outbreaks ('Infreq'), frequent recurrent outbreak with episodic treatment ('Freq (episodic T)'), and 'all other' (including the primary outbreak, frequent recurrent outbreak on CST, mild to moderate complications, and severe complications).

Effective intervention strategies for neonatal herpes are urgently needed.^{5,65} Neonatal herpes causes more infant deaths in the United States than congenital syphilis or HIV.⁶⁶ While on average genital HSV infection leads to a relatively modest reduction in the quality of life of an adult,^{18,55} neonatal herpes infection is notable for its devastating outcomes. The 2021 *Sexually Transmitted Infections Treatment Guidelines* recommend asking all pregnant women about their history of genital herpes or genital symptoms concerning HSV infection

and performing cesarean delivery for women with genital herpetic lesions at the onset of labour to reduce the risk of neonatal HSV infection.⁶⁷ Interventions that aim to improve healthcare providers' adherence to this guideline may be useful. Neonatal herpes is only notifiable in some jurisdictions and there is no national-level surveillance system.⁶⁸ There would be benefits in gathering more robust national-level data on neonatal herpes, which would aid in the evaluation and development of prevention strategies.^{65,68}

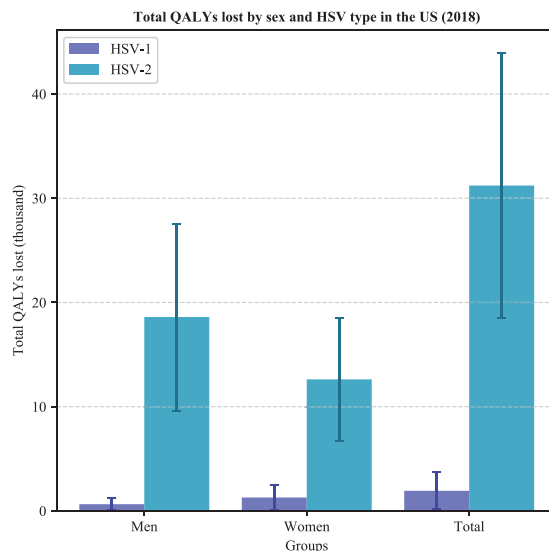


Fig. 5: The total lifetime number of QALYs lost due to genital HSV infections acquired in 2018 among people aged 18–49 years in the U.S. population, by sex and HSV type. Error bars represent one standard deviation uncertainty below and above point estimates, which were approximated as one-fourth of the 95% uncertainty ranges. The precise results are reported in the Supplement (Suppl. Table S3).

Our input parameters were informed by literature reviews, but where data was limited, values for input parameters were more uncertain. Our estimates were sensitive to assumptions around the psychosocial disutility associated with GH. Few studies were found

that quantified the health losses due to psychosocial effects among people diagnosed with HSV. A systematic review on health-related quality of life for people with genital herpes identified 19 studies reporting primary quality-of-life data, 18 of which used diverse health questionnaires to evaluate psychosocial effects, such as the Recurrent Genital Herpes Quality of Life Questionnaire (RGHQoL), the General Health Questionnaire (GHQ), and the Hospital Anxiety and Depression Questionnaire (HADQ).⁶⁹ Only one study conducted in 2005 among 39 patients with recurrent GH evaluated quality-of-life impacts using direct valuation that could be used to generate utility weights, which is the study we adopted to inform model parameterization.^{18,69} Results indicated that the value of psychosocial disutility significantly influences the total QALYs lost. While studies report that people on CST or people with less frequent recurrences experience less psychosocial distress, there is a lack of robust quantitative estimates of these effects using standard health utility metrics. Although we identified several studies using health questionnaires to measure psychosocial impacts, there is no conversion equation available to convert questionnaire scores to standard utility values.¹⁸ More empirical data are needed to improve the precision of this estimate. Such data could also have important implications regarding the need for and potential benefits of counselling strategies, as a way to reduce the overall health losses for patients with GH.

In addition to the psychosocial factor, the symptomatic recurrence of GH is another leading factor contributing to the total QALYs lost associated with per

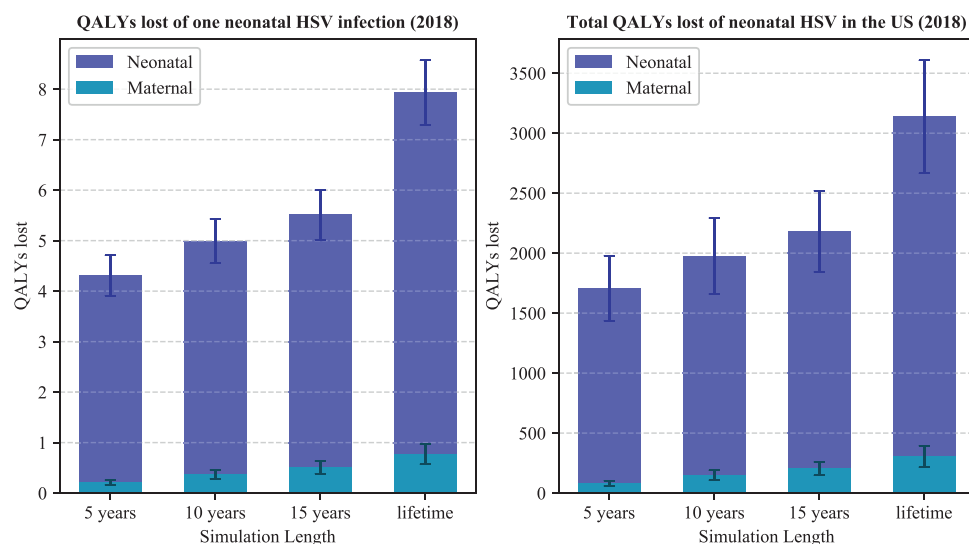


Fig. 6: Average number of QALYs lost per neonatal HSV infection acquired in 2018 in the US population (left panel) and total number of QALYs lost due to neonatal HSV acquired in 2018 in the US population (right panel). The simulation length represents the length of time considered in the analysis after the first year of neonatal disease onset. Lifetime QALYs lost were used in the main analysis. Error bars represent one standard deviation uncertainty below and above point estimates, which were approximated as one-fourth of the 95% uncertainty ranges. The precise results are reported in the Supplement (Suppl. Table S4).

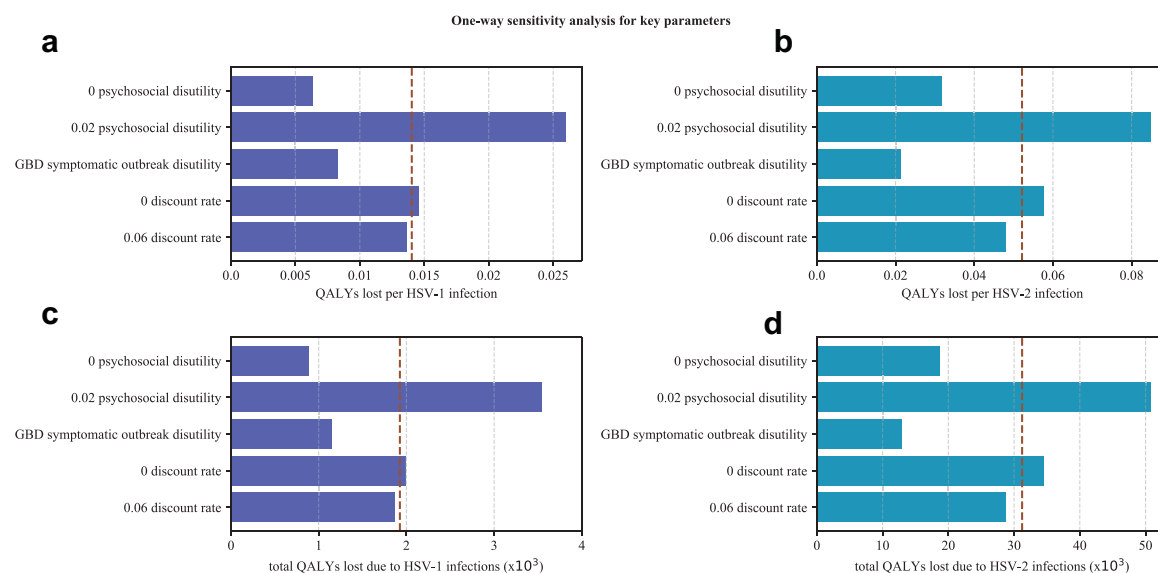


Fig. 7: One-way sensitivity analyses on key parameters. The vertical dark red dashed lines represent base case outcomes. Panel (a) and panel (b) display the lifetime QALYs lost per genital HSV-1 and HSV-2 infection in the US among people aged 18–49 years, respectively. Panel (c) shows the total QALYs lost associated with genital HSV-1 in the US among people aged 18–49 years, and panel (d) shows the total QALYs lost associated with HSV-2 infections in the US among people aged 18–49 years. In the base case scenario, drawing on a prior study by Fisman,¹⁸ we modelled the psychosocial disutility following diagnosis as a 1% reduction initially. We assumed a one-time reduction in disutility after the year of diagnosis and the disutility remains constant thereafter and decreases to zero on the year the person stops experiencing symptomatic GH recurrences. We assigned the proportion of HSV-1 infections that were genital as 20%,¹ the utility loss from a symptomatic outbreak as 0.15,¹⁸ and the discount rate as 0.03 per year. The utility estimate for a symptomatic outbreak was based on Fisman's study on a small sample of patients ($n = 39$) from a genital herpes support group;¹⁸ these patients may over-report their utility loss associated with an outbreak. Therefore, we also considered the utility loss estimate from the GBD study,⁵⁵ which is a non-STI-specific utility estimate for a symptomatic outbreak. The GBD symptomatic outbreak disutilities were 0.05 per primary symptomatic outbreak and 0.006 per recurrent symptomatic outbreak.

GH infection, especially for HSV-2. However, we found limited evidence to inform parameterization of the recurrence rate by year after the primary episode.⁵⁷ We assumed a linear decrease over time in the number of recurrences for adults with HSV infections. We used this assumption because the main goal of our study was to estimate the average number of QALYs lost per infection. Another evidence gap pertains to the interactions between HSV-1 and HSV-2 co-infections. Limited evidence has suggested prior infection with HSV-1 may alter the clinical manifestations of a subsequent HSV-2 infection.¹⁷ However, given the high prevalence of oral HSV-1, we assumed the possible effects of prevalent HSV-1 on HSV-2 were embedded in the current estimates of recurrences and sequelae. Thirdly, we did not stratify the burden of GH by race/ethnicity due to the lack of available data.

For neonatal HSV infection, we did not account for the short-term health loss during the acute infection period (i.e., utility losses due to SEM/CNS/disseminated disease within the first month after disease onset) because no disutility estimates were found in identified studies.^{12,43,47,52,70–72} The disability weights of the long-term neurological sequelae due to neonatal herpes were also outdated. We identified six modelling studies published

between 2004 and 2021 assessing the intervention effectiveness of neonatal herpes; all of them adopted utility estimates from studies before 2000.^{12,47,52,70–72} The lack of quantitative data on long-term outcomes among neonatal herpes survivors limits more nuanced assessment of the burden of neonatal herpes, and it is also a limitation shared by some other conditions afflicting neonates, such as congenital syphilis.

Our study made several conservative assumptions that may underestimate the true burden. We considered only the direct effects of genital HSV infections on the losses of QALY. A study on HSV-2 and HIV interactions indicated that if prevalent genital HSV-2 infection increases HIV acquisition risk, 25.1% (95% UI 20.9–30.8%) of the incident, sexually-acquired HIV infections could be attributed to HSV-2 between 2019–2028 in the Americas region.⁷³ We did not include encephalitis as an adverse outcome into our model. While encephalitis is a rare but devastating complication associated with HSV-1 infections, there is scarce evidence of genital HSV-1 infections contributing to encephalitis development.⁷⁴ Thirdly, we adopted a wide uncertainty range for the proportion of HSV-1 infections to be genital, from 5% to 75%, to reflect the evidence gap in HSV-1 epidemiology. Compared to Spicknall et al. estimated, which we

adopted to inform the incidence of HSV-2 in the US, Ayoub et al. estimated a similar, albeit slightly higher HSV-2 incidence in 2018.^{6,7}

Our study provides estimates of the lifetime number of QALYs lost due to GH infections acquired in 2018 among people 18–49 years old in the United States. Estimating the burden of GH using QALYs was challenging due to limited data on key model parameters, including the incidence of genital HSV-1, the psychosocial impacts due to HSV diagnosis and recurrences, and the disutility associated with a symptomatic GH outbreak. To address these challenges, we performed a probabilistic sensitivity analysis to estimate 95% uncertainty intervals of QALYs lost in the main analysis and conducted additional one-way sensitivity analyses to examine the impact of key parameters. Our study quantifies the public health burden of GH in the context of the United States. The estimates can be used to inform future research regarding the health burden of GH and interventions, such as studies that evaluate the impact and cost-effectiveness of HSV preventive interventions.

Contributors

MMR, JAS, HWC, and TLG conceived the study idea. SY developed the analytic framework and parameterized the models with guidance from MMR, RY, JAS, STE, HWC, TLG, KKH, KL, YL, and AAB. SY conducted analyses and produced results. SY, MMR, RY, JAS, STE, HWC, TLG, KH, AAB, KL, and YL interpreted the study findings. SY wrote the first draft of the manuscript with MMR, which was edited by HWC, STE, JAS, RY, TLG, KKH, AAB, KL, and YL. All authors (SY, MMR, RY, JAS, STE, HWC, TLG, KKH, AAB, KL, YL) read and met the ICMJE criteria for authorship and agree with the results and conclusions. SY and MMR had access to the data in the study and SY and MMR had final responsibility for the decision to submit for publication.

Data sharing statement

All data used in this study are available online and can be accessed through the sources listed in Tables 2–4. The analytic code (Python 3.8) is available as open access on GitHub: <https://github.com/shiying88/QALYs-Lost-due-to-Genital-Herpes>.

Declaration of interests

MMR reports contract for work on cervical cancer elimination modeling in South Africa, not related to the manuscript, from World Health Organization, travel and lodging support for attending a research project meeting at the World Health Organization headquarters in Geneva, Switzerland, not related to the manuscript, from World Health Organization. YL reports postdoctoral fellowship stipend (2020–2022) from Harvard T. H. Chan School of Public Health, not related to the manuscript. SY reports doctoral stipend from New York University Grossman School of Medicine, not related to the manuscript. All other authors report no potential conflicts.

Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.lana.2023.100427>.

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